Diwali Sales Data Analysis

1.Project Overview :

This project analyzes Diwali Sales data analyisi using transactional data from 11,251 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and Top sales trends to guide strategic business decisions.

2. Dataset Summary :

- Rows: 11251
- Columns: 18
- Key Features:
- Customer demographics (Age, Gender, State, Marital Status,)
- Sales details (Product Category, Amount, Orders, Occupation)
- Missing Data: 12 values in Amount column

3. Exploratory Data Analysis using Python :

We began with data preparation and cleaning in Python:

- **Data Loading**: Imported the dataset using pandas.
- **Initial Exploration**: Used df.info() to check structure and .describe() for summary statistics.

	User_ID	Age	Marital_Status	Orders	Amount
count	1.123900e+04	11239.000000	11239.000000	11239.000000	11239.000000
mean	1.003004e+06	35.410357	0.420055	2.489634	9453.610553
std	1.716039e+03	12.753866	0.493589	1.114967	5222.355168
min	1.000001e+06	12.000000	0.000000	1.000000	188.000000
25%	1.001492e+06	27.000000	0.000000	2.000000	5443.000000
50%	1.003064e+06	33.000000	0.000000	2.000000	8109.000000
75%	1.004426e+06	43.000000	1.000000	3.000000	12675.000000
max	1.006040e+06	92.000000	1.000000	4.000000	23952.000000

- **Missing Data Handling**: Checked for null values and Drop missing values in the Amount column.

- **Column Standardization**: Renamed columns to snake case for better readability and documentation

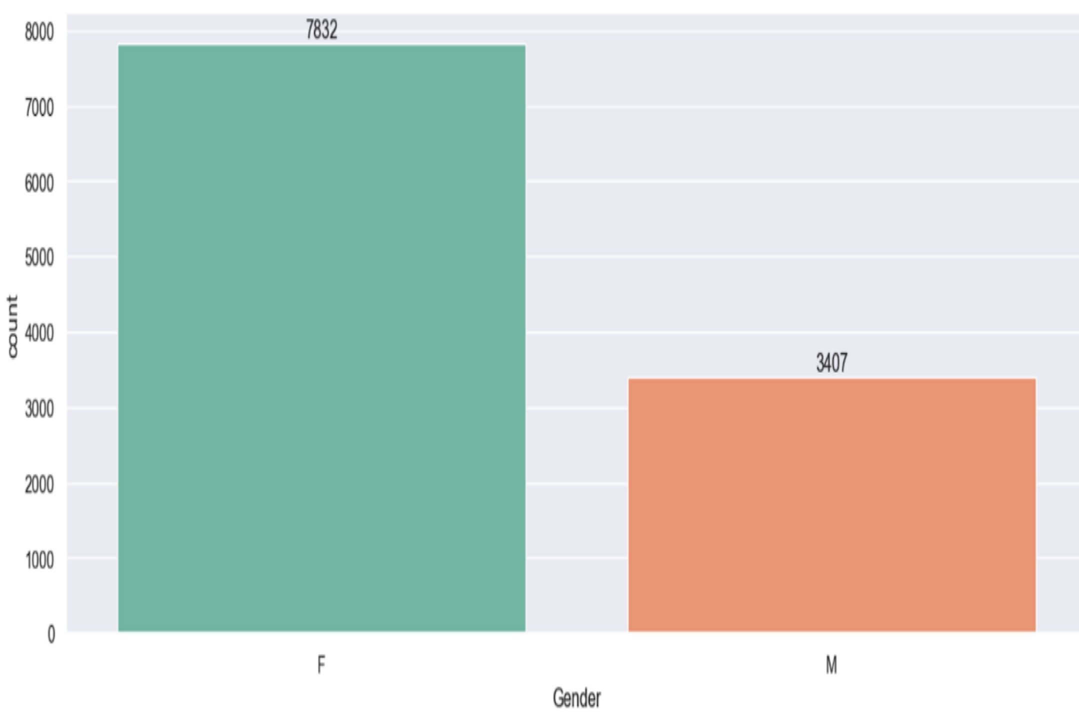
- **Feature Engineering:**
  - Changed Amount column datatype by int32.
  - Renamed Shadi column to Marital Status .

- **Database Integration**: Connected Python pandas to the Seaborn and matplotlib after Imported in jupyter notebook.
  cleaned DataFrame for analysis and visualisation.

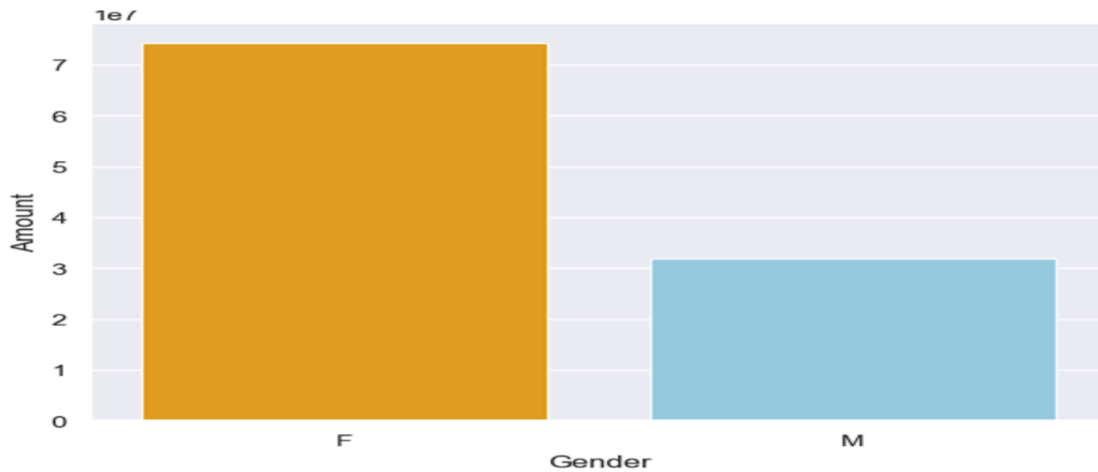
4.Data Analysis using pandas and seaborn (Business Transactions):

We performed structured analysis in PostgreSQL to answer key business questions:

1. **buyers by Gender -** Compared total count generated by male vs. female customers.

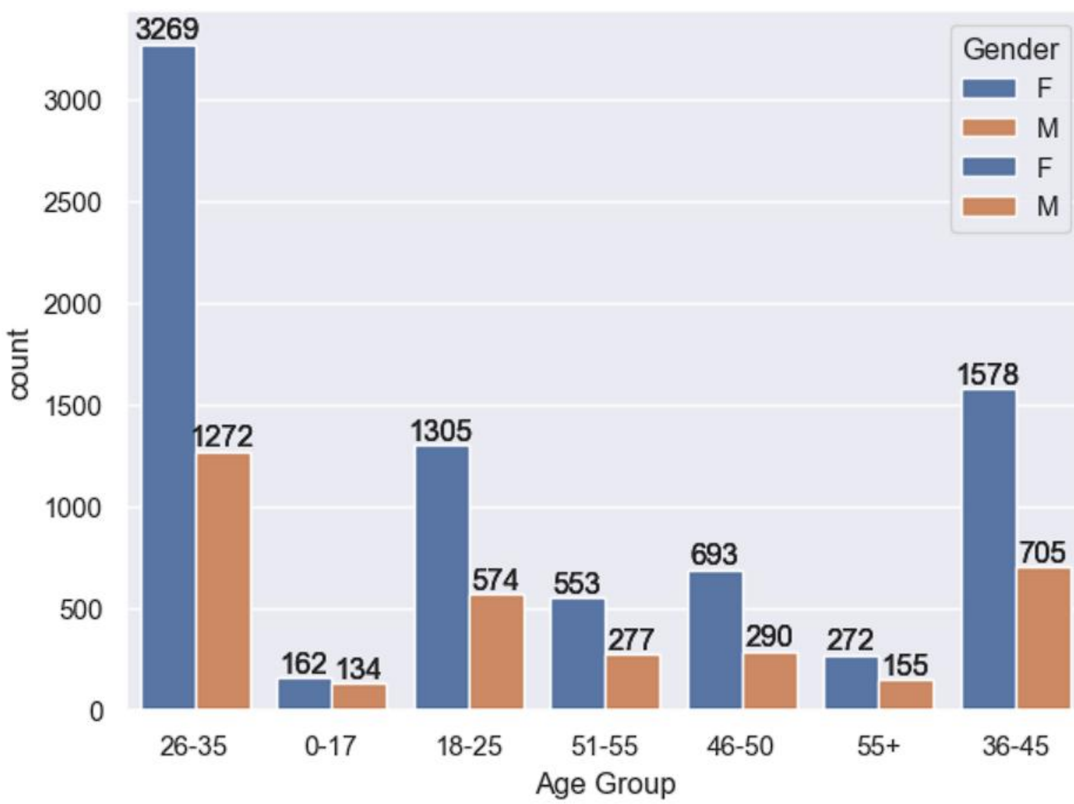


2. **Amount Purchasing by Gender -** Compared total purchasing amount of female are greater than male.



we can see that most of the buyers are females and even the purchasing power of females are greater than men

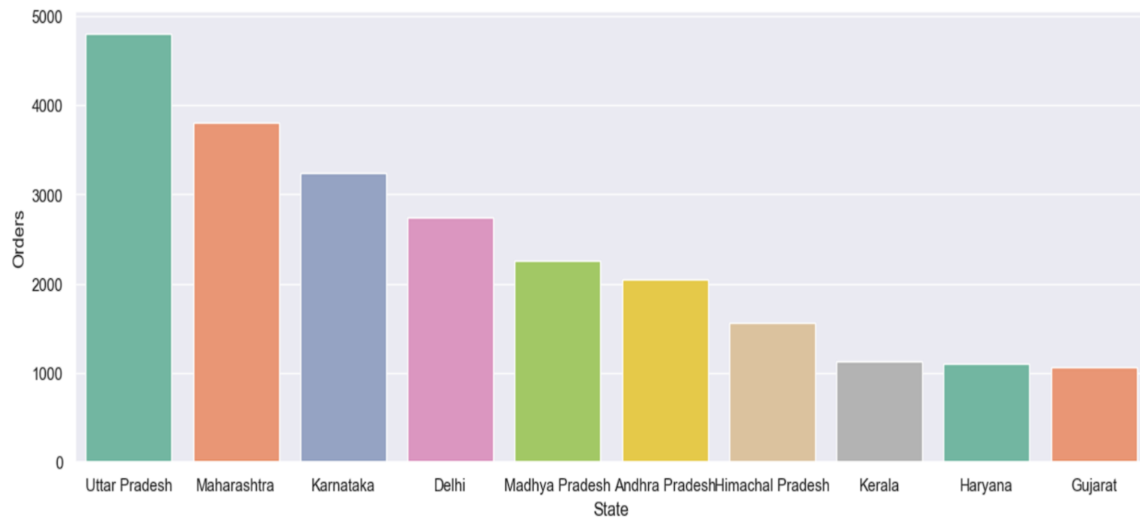
3. **Buyers by Age Group -** Compared total count generated by Age Group customers.



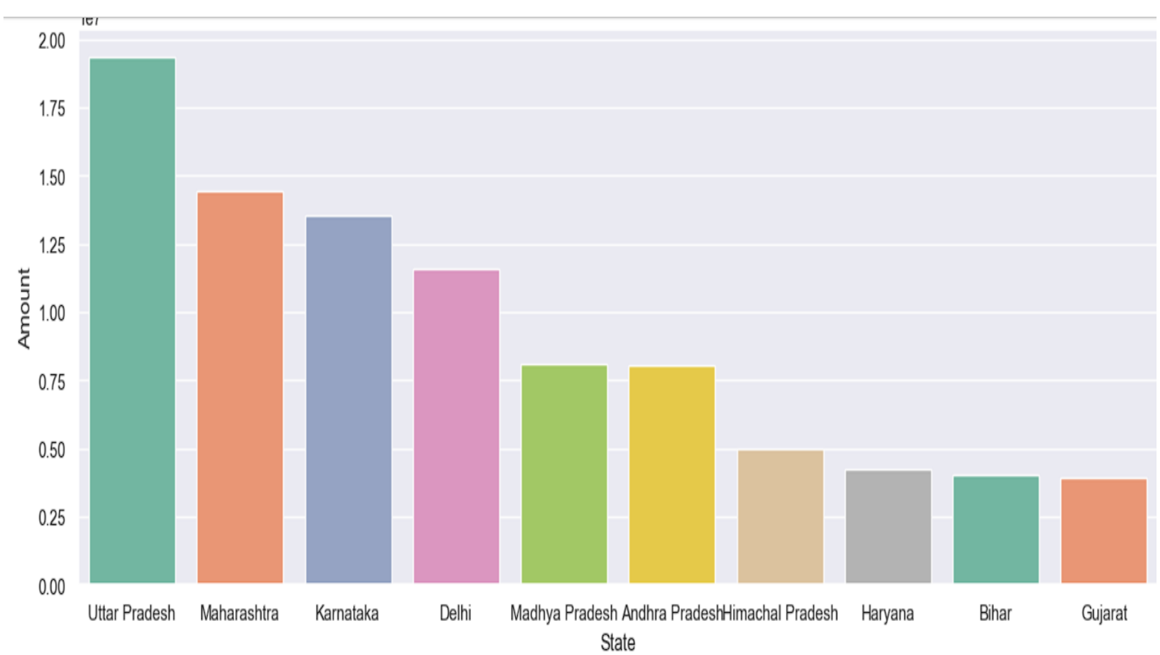
we can see that most of the buyers are of the age group between 26 -35 yrs female

4. **Orders by States -** Total number of orders and sales from the top 10 states.

Top orders by state



Top sales by state

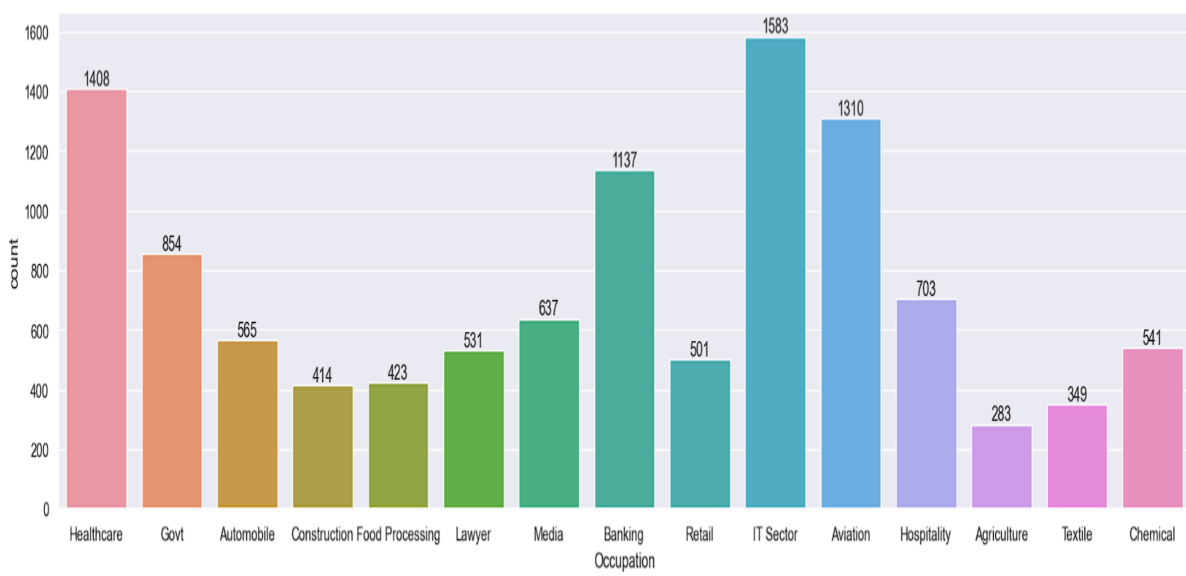


We can see that most of the orders and sales are from the Uttar Pradesh, Maharashtra, Karnataka, Delhi, Madhya Pradesh.

5. **Sales by Marital status -** Compared total count generated by Marital status with gender.

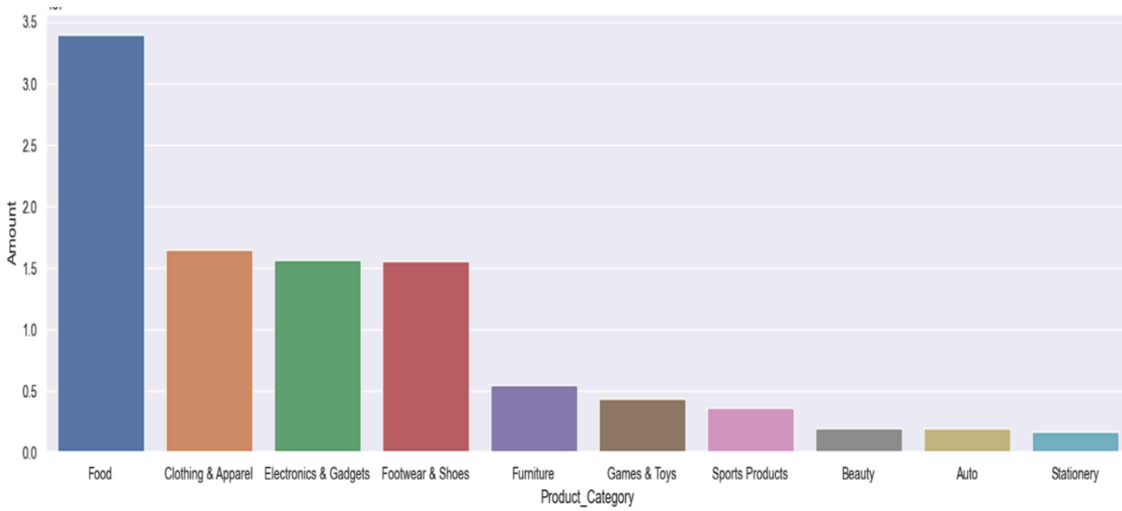
we can see that most of the buyers are married (women) and they have high purchasing power

6. **Analyzed sales by occupation -** Most sales based on customer occupation to Identify high purchasing professions



most of the buyers are working in IT, Healthcare and Aviation sector

7. **Sales top product -** Listed the most sold products.



most of the sold products are from Food, Clothing and Electronics category

5. Conclusion :

Married women age group 26-35 yrs from UP, Maharastra and Karnataka working in IT, Healthcare and Aviation are more likely to buy products from Food, Clothing and Electronics category